Smart Food Spoilage Detection System

Using NodeMCU, TCS34725 RGB Sensor, and MQ135 Gas Sensor

Team

Team Members:

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Academic Year

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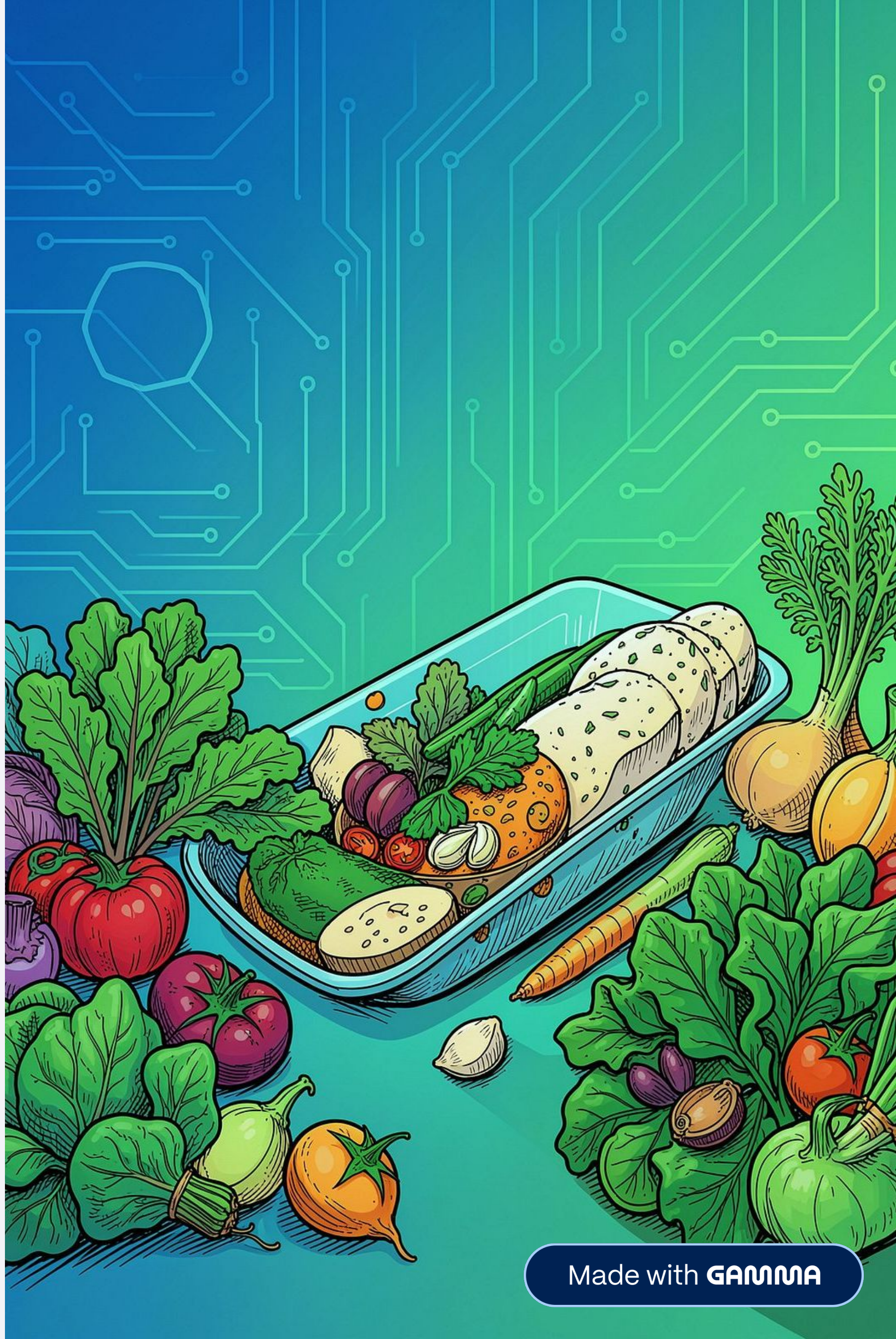


Problem Statement

Health Risks

Food spoilage causes health issues when contaminated food is consumed.

Inaccuracy

Manual inspection is subjective and often misses early spoilage.

Need

Requirement for a low-cost, automated smart monitoring system.

Waste

Global food waste is increasing; early detection can reduce losses.

Objectives

1

Automatic Detection

Detect spoiled food automatically without manual checks.

2

Gas Measurement

Measure harmful gases with MQ135 to identify spoilage by-products.

3

Color Detection

Detect color changes with TCS34725 to spot visual spoilage cues.

4

Data Transmission

Send sensor data via NodeMCU for processing and alerts.

5

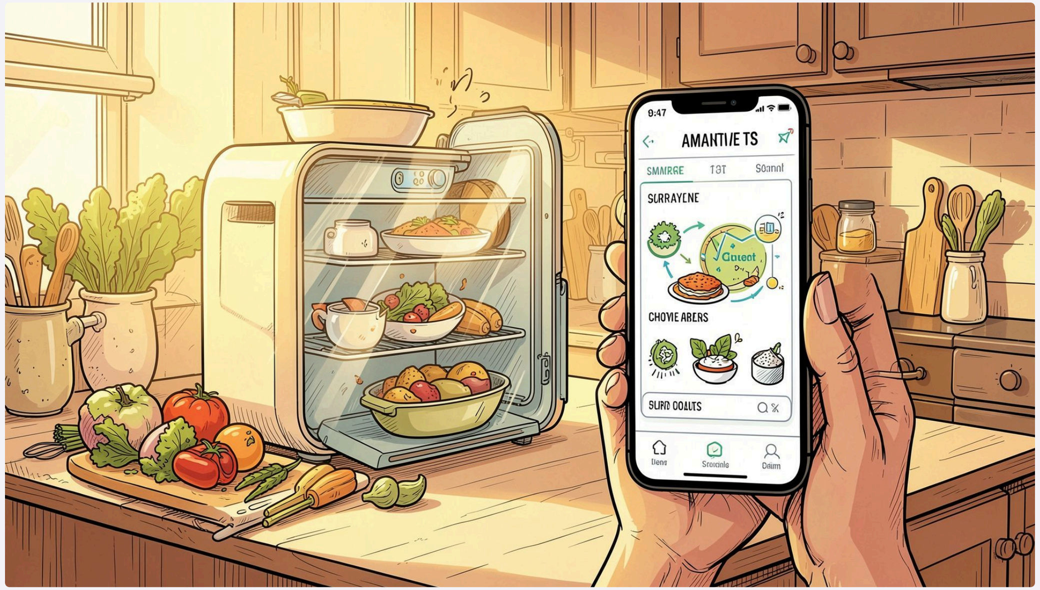
Early Indication

Provide timely Fresh/Spoiled indication to prevent consumption and waste.

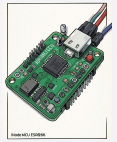
Introduction

Food spoilage is caused by microbial growth, chemical reactions, and environmental exposure. Monitoring preserves safety and reduces waste.

IoT-based food quality monitoring enables continuous, remote checks, combining sensors, microcontrollers, and network connectivity for real-time alerts.

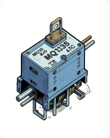


Components Used



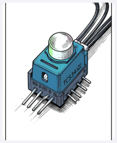
NodeMCU ESP8266

Wi-Fi microcontroller for data transmission and processing.



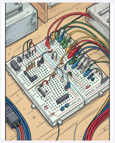
MQ135 Gas Sensor

Detects ammonia, NOx, benzene—useful for spoilage-related gases.



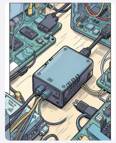
TCS34725 RGB Sensor

Measures red/green/blue values to detect color changes in food.



Breadboard & Wires

Prototyping hardware for connections and signal routing.



Power Supply

Stable 5V supply for NodeMCU and sensors.

Hardware Description & Working Principles

NodeMCU

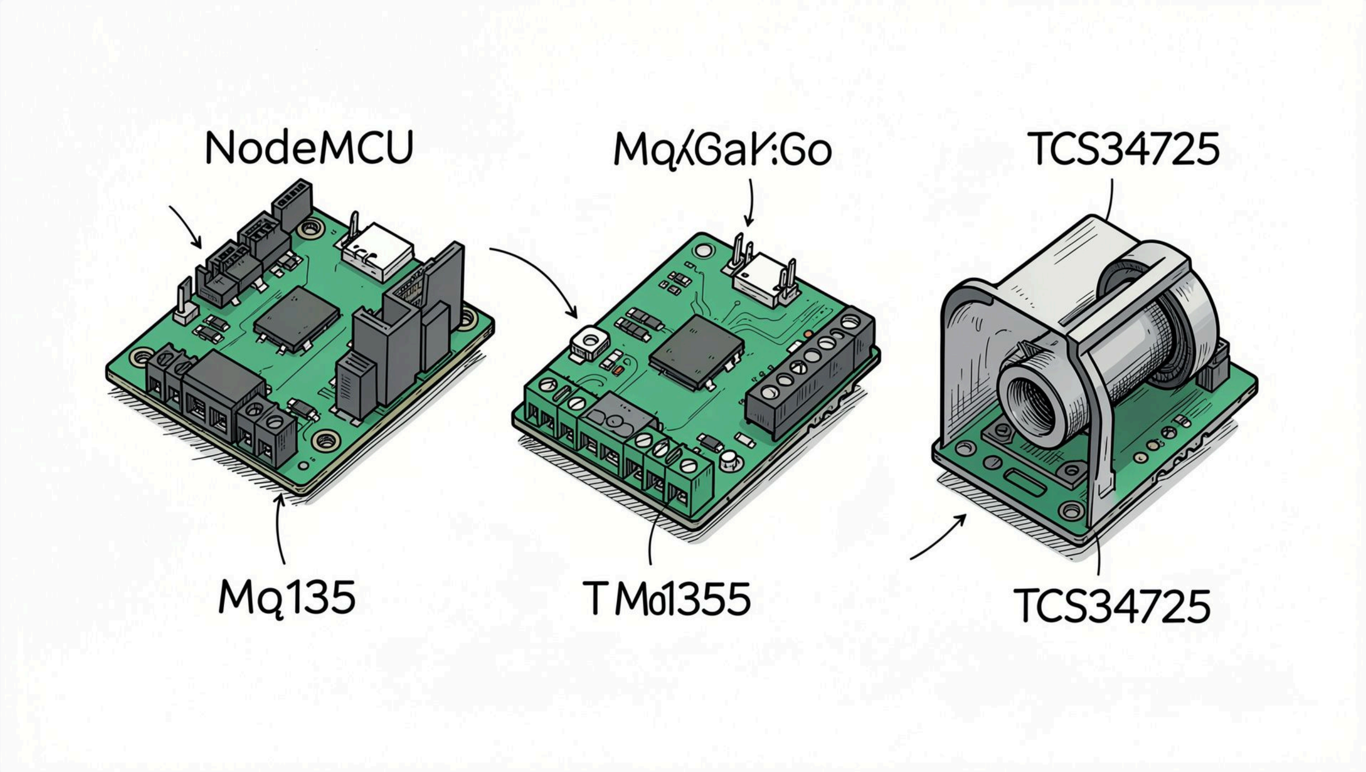
ESP8266-based board: reads sensor data, runs logic, and sends updates via Wi-Fi.

MQ135

Semiconductor gas sensor whose resistance changes with target gas concentration; calibrated to approximate ppm.

TCS34725

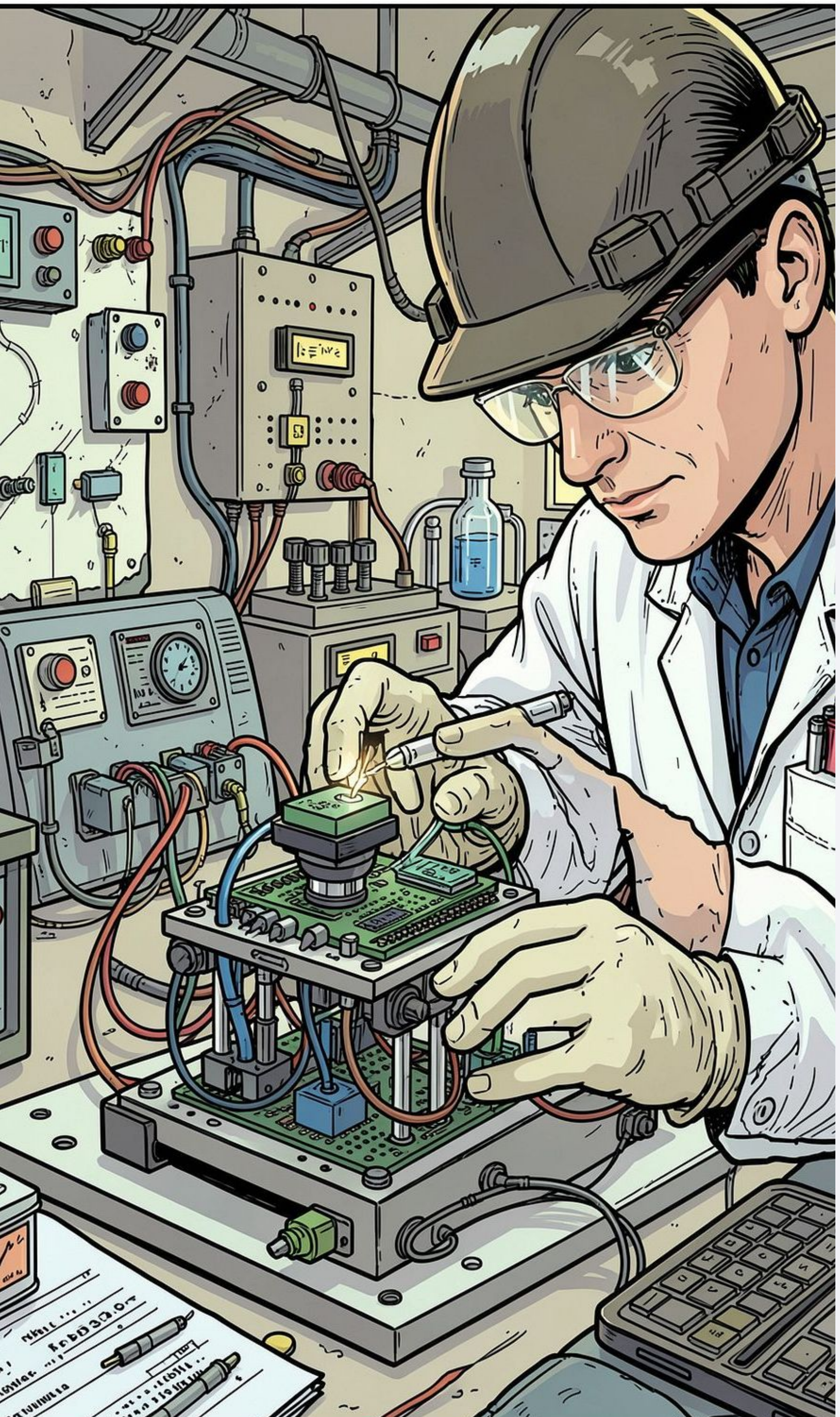
RGB color sensor with IR filter and onboard LED; outputs calibrated R,G,B readings for color analysis.



System Architecture

- Food Sample
- Color Sensor
- Gas Sensor
- NodeMCU

Modular chain: sensors capture environmental and visual cues → NodeMCU aggregates and evaluates against thresholds → outputs status and uploads data.



Working Principle (Step-by-step)

01

1. Place Sample

Food sample positioned under sensors for stable readings.

02

2. Sensor Readings

TCS34725 captures RGB values; MQ135 measures gas concentration.

03

3. Data Processing

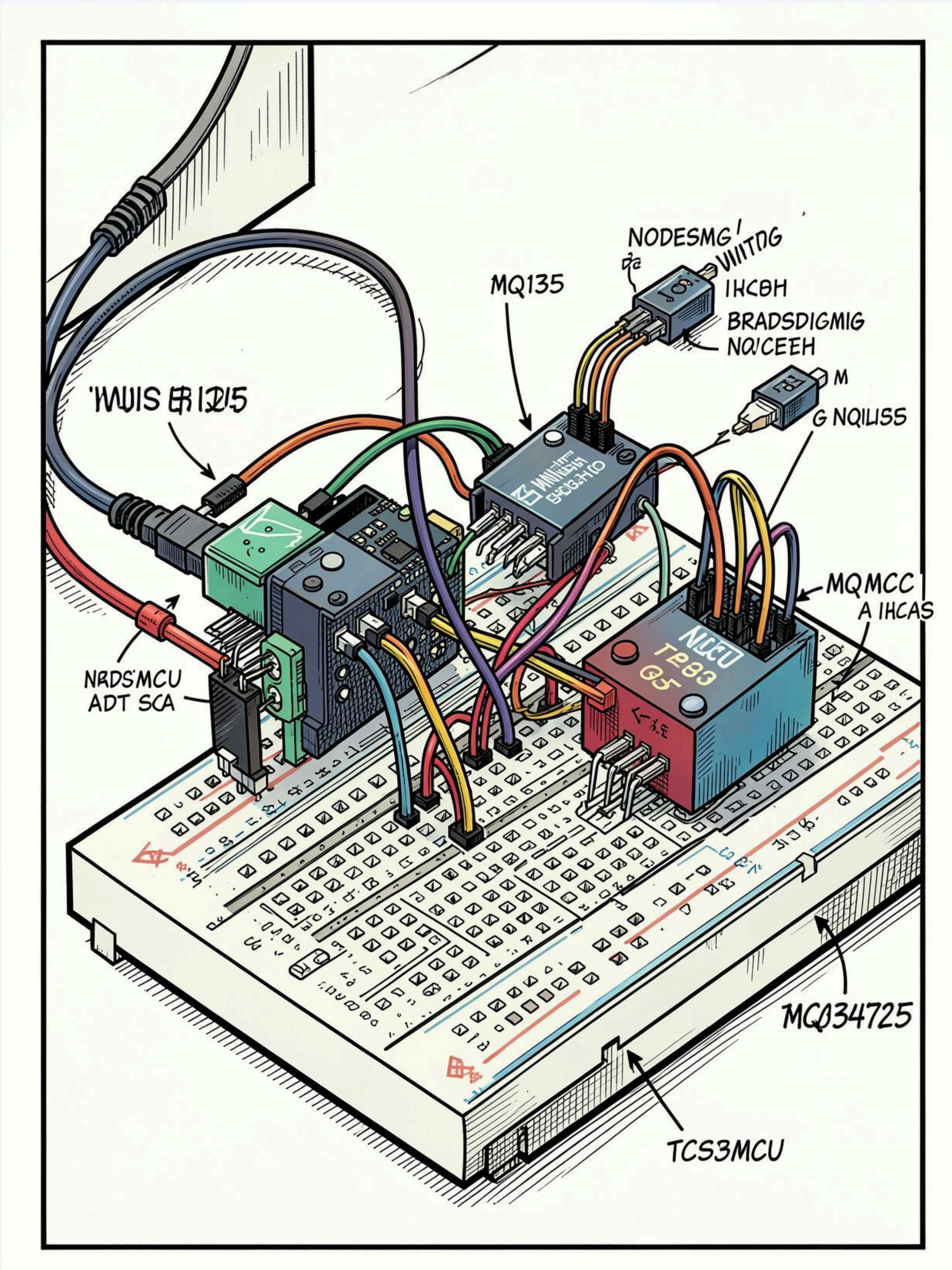
NodeMCU filters, converts, and compares values to thresholds.

04

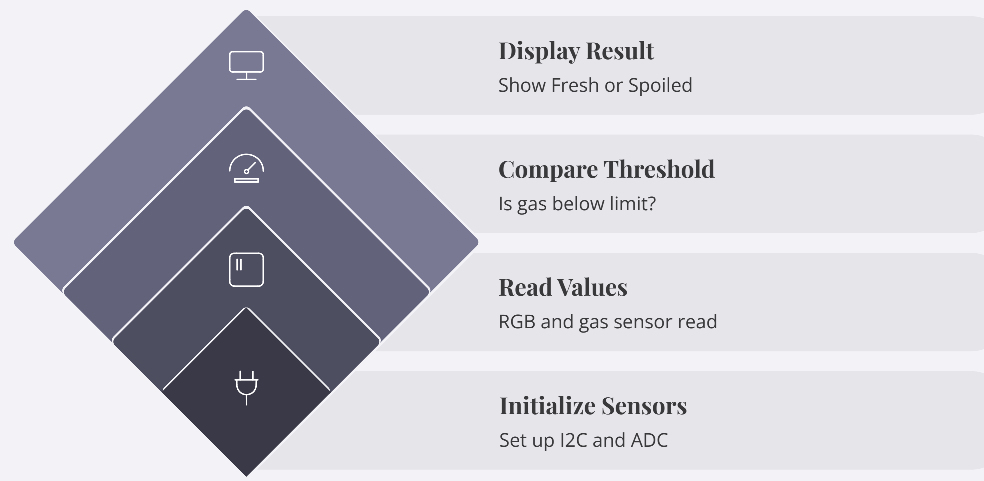
4. Decision & Output

If values exceed thresholds → Spoiled; else → Fresh. Results displayed and sent via Wi-Fi.

Circuit Diagram & Flowchart



GPIO: I2C for TCS34725 (SDA, SCL), analog/digital read for MQ135, power and ground shared.



Looped monitoring with threshold-based decision and continuous updates.

Results, Advantages & Next Steps

Advantages

Low cost, easy to build, IoT-enabled, real-time monitoring, portable, reduces food waste.

Applications

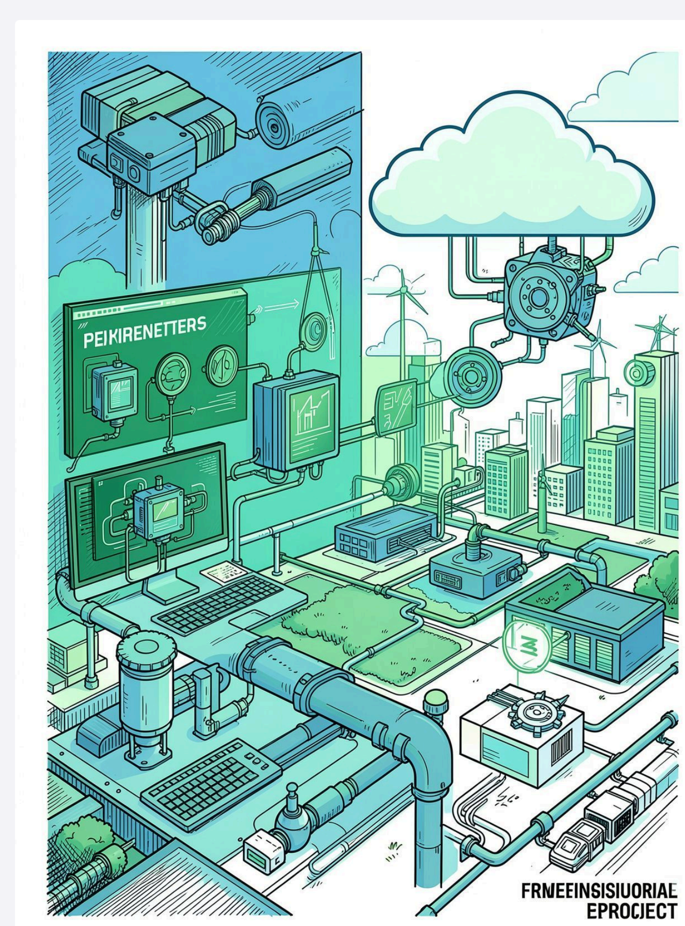
Refrigerators, food industries, supermarkets, hotels, cold storage, smart kitchens.

Future Scope

AI prediction, mobile app, cloud storage, camera integration, machine learning, multi-food detection.

Conclusion

The system fuses gas and color sensing to reliably indicate spoilage, improving food safety and reducing waste.



Sample Results (table)

Sample	RGB / MQ135	Status
Fresh Apple	220,45,40 / 85	Fresh
Spoiled Apple	135,60,40 / 310	Spoiled

Charts: RGB shifts and gas ppm trends illustrate clear separation between Fresh and Spoiled states.